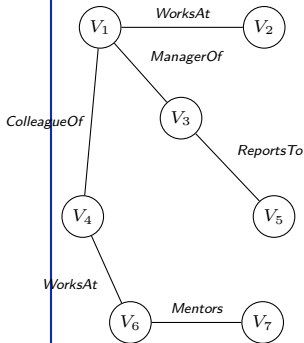


1. Original Graph

$G = (V, E, X, R)$



V : entities (nodes)
 E : relations (edges)
 X : node attributes
 R : relation information

2. Policy-Guided Sensitivity Selection



Protected entity selection

Identify entities to be pseudonymized (policy-guided).



Policy-designated sensitive relation selection

Identify relations whose original labels require protection.



Protection policy

Define rules and scope of transformations (e.g., relation protection, rewiring variant).

3. TAFP Protection Layer (Offline)

Offline pre-indexing graph protection

Core TAFP contribution



3A. Deterministic entity pseudonymization

Selected entities are replaced with deterministic pseudonyms (reproducible and auditable).



3B. Original sensitive relation-label replacement

Original labels of selected sensitive relations are replaced with a common protected marker.

Original label → **PRIVATE_RELATION**



3C. Exact degree-preserving rewiring

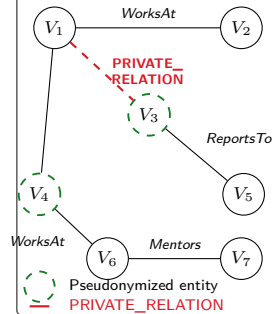
Edges are rewired using exact degree-preserving transformations (for designated degree-preserving variants).

Guarantees (design goals)

- Sensitive relation-label exposure reduced
- Node degrees preserved (for designated variants)

4. Outputs

Protected Graph G'



Audit Log A



- Edits (e.g., entities, relations, edges)
- Seeds and parameters
- Realized transformations
- Integrity checks
- Runtime and manifests

5. Downstream GraphRAG Pipeline



5A. Indexing

Protected graph G' is indexed.



5B. Retrieval

Topology-aware retrieval over G' .



5C. Retrieval-Grounded Generation / Relation-QA

LLM generates answers grounded on retrieved subgraphs.

Node types



Original entity



Pseudonymized entity

Edge types

— Regular (non-sensitive) relation

- - Sensitive relation (original)

— Protected relation (PRIVATE_RELATION)

Key

G : original graph

G' : protected graph after TAFP

A : audit log (evidence and metadata)

Notation in system

TAFP is an offline, pre-indexing protection layer that operates before any GraphRAG indexing or retrieval.